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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	SyncEra
Selected Domain	Domain 1 — The Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	Proposed primary: SDG 4 (Quality Education); supporting: SDG 8 (Decent Work and Economic Growth).
Project / Solution Name	Skillogy
Name (Leader)	HAMMADOUCHE Manar
Email & Phone (Leader)	ERA2509065@xmu.edu.my +601121047468
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

The core challenge facing job seekers and workforce institutions today stems from a fundamental contradiction: people are making high-stakes career decisions based on incomplete self-perception and rapidly changing, fragmented employment market information. Working adults, fresh graduates, and career changers rely on static CVs and self-reported surveys that omit transferable skills, misrepresent proficiency, and ignore behavioral learning contexts like motivation and persistence, leading to poorly targeted recommendations that waste time and money. This misallocation of talent carries significant economic consequences; for instance, the Department of Statistics Malaysia (DOSM) reported in Q2 2026 that approximately 35% of tertiary-educated Malaysians were employed in semi- or low-skilled jobs, signaling a severe structural mismatch between qualifications and actual job requirements. Simultaneously, HR/L&D teams, universities, TVET providers, and career centers lack a consistent, evidence-linked mechanism to measure individual skill gaps and align educational programs with real-time industry demand. Directly addressing UN Sustainable Development Goals 4 (Quality Education) and 8 (Decent Work and Economic Growth), Skillogy bridges the gap between individual capability awareness and real-world market requirements, transforming static, unreliable profile matching into an adaptive, evidence-grounded career navigation pathway.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Skillogy combines skills intelligence with psychology-informed reflection to support human-centred career navigation. It connects dated market signals with a person's reviewed skills, potential gaps and practical next steps. The current Career Radar prototype uses rule-based scoring and synthetic demo data. Its Moderate and Fast AI-transformation scenarios illustrate different possible rates of change; they are not forecasts of job stability. An optional, off-by-default Tencent TokenHub/MiniMax integration can extract skill and experience claims from synthetic CV profile data, but CV extraction and review are not yet part of the main user flow. We aim to let users inspect the source text, correct extracted claims and compare confirmed skills with approved role requirements and Malaysian labour-market data. A transparent recommendation layer would identify skill gaps, suggest learning steps and explain the evidence behind each recommendation. Optional reflection on motivation, persistence and work habits could help tailor learning support; it does not currently affect career scores or recommendations. A future consistency-check feature could ask users to clarify differences between CV claims and their examples. It is not implemented or validated, and a mismatch would not prove dishonesty. Nature shapes our two-sided vision: **bamboo's interconnected rhizomes** represent shared support for individual resilience and growth, while **Boston ivy's** adaptive growth represents organisations adjusting talent development to changing needs. Future privacy-protected, aggregated insights could help HR/L&D teams, universities, TVET providers and career centres identify shared skill gaps. Built with React/Vite and FastAPI, the prototype uses SQLite locally, with optional PostgreSQL configuration. Skillogy aims to support SDG 4 through relevant learning and SDG 8 through pathways towards decent work.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Skillogy's architecture separates deterministic, rule-based recommendation logic from optional, supplementary GenAI tasks. The system prioritizes transparent matching over black-box AI scoring, using synthetic baseline data and modular storage layer interfaces.

- Frontend: Built with React, TypeScript, and Vite, delivering a responsive single-page application for role exploration, skill gap visualization, and user self-reflection.
- Backend and API: Powered by Python, FastAPI, and Pydantic for typed data validation across /api/v1 RESTful endpoints.
- Recommendation Engine: Employs deterministic, rule-based matching algorithms for role-scoring and skill-gap calculations. AI/LLM processing is strictly decoupled from career scoring: an optional, off-by-default server-side LLM task extracts and normalizes skill claims from user-provided text/CVs to display supporting source context for user review, but does not calculate or modify suitability scores.
- Data Taxonomy & Baseline Limits: Utilizes versioned market snapshots, skill catalogues, and approved role taxonomies. The application defaults to synthetic demo data; national statistics (such as DOSM occupation groups) provide broad macro-level context rather than direct proof of fit for specific job roles.
- Storage & Infrastructure: SQLite serves as the zero-config local database default, with an optional PostgreSQL configuration for hosted/multi-user environments. A Vercel preview API proxy is included for interface testing, while final production deployments remain host-agnostic.
- AI Integration: Includes an optional, server-side integration for Tencent TokenHub / MiniMax dedicated solely to unstructured text extraction on synthetic profile data.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We combined human-centred design thinking with iterative validation. Reviewing career trends and representative scenarios revealed shared challenges for individuals and organisations: incomplete skills evidence, fragmented labour-market information, and insufficient personalised learning guidance.

We defined three goals: trustworthy profiles, explainable career guidance, and accessible experiences. We conceived an integrated platform instead of isolated job matching. Bamboo roots symbolise interconnected knowledge supporting individual growth; Boston ivy represents organisational adaptability. Our “skills + psychology” concept includes optional reflection questionnaires and MBTI preferences for self-exploration, without using these inputs to calculate suitability scores.

We first prototyped Career Radar using React, FastAPI, synthetic data, and deterministic skill matching. Modular development preserved scoring transparency and separated experimental AI components.

Following iterations, developers performed internal walkthroughs to examine navigation, skill-gap explanations, and learning actions. Formal external testing remains planned: in Phase 2, ten participants will assess whether they can independently explain recommendations, identify skill gaps, and select learning steps. We target at least 80% unaided completion.

Staged deployment will require API validation, database preparation, privacy safeguards, informed consent, and controlled testing before wider use.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 4.4:** increase youth and adults with relevant skills for employment, decent jobs and entrepreneurship. **SDG 8.5:** support productive employment and decent work. Skillogy aims to contribute through clearer skill-gap and learning guidance; it cannot guarantee employment outcomes.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	DOSM reported **35.2%** skill-related underemployment among employed people with tertiary education in Q2 2026. This is the national problem indicator, not a baseline for Skillogy users. The prototype's user-outcome baseline has not yet been measured; collect it before participants use the prototype.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Evidence-informed pathway success rate:** percentage of pilot participants who, without facilitator help, can identify a recommended role's evidence, state its main skill gap and choose a relevant next step. Measure the same task before and after using Skillogy.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	At least **8 of 10 participants (80%)** complete the pathway task unaided after using the prototype. Report the pre-use result alongside it; treat this as a small usability result, not evidence of reduced national underemployment.

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
DOSM JMI + Employment Statistics, Q2 2026	dataset	DOSM reuse/attribution; verify terms
DOSM reuse/attribution; verify terms	dataset	Publisher reuse terms; verify

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24/10/2026